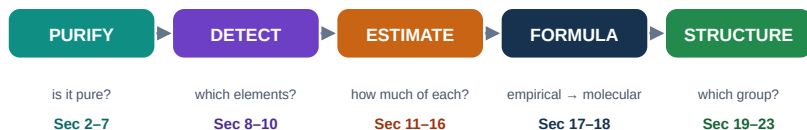
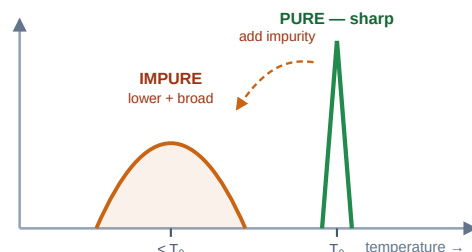


1 The Whole Chapter in One Picture

OVERVIEW



Every question in this chapter sits at exactly one of these five stages.



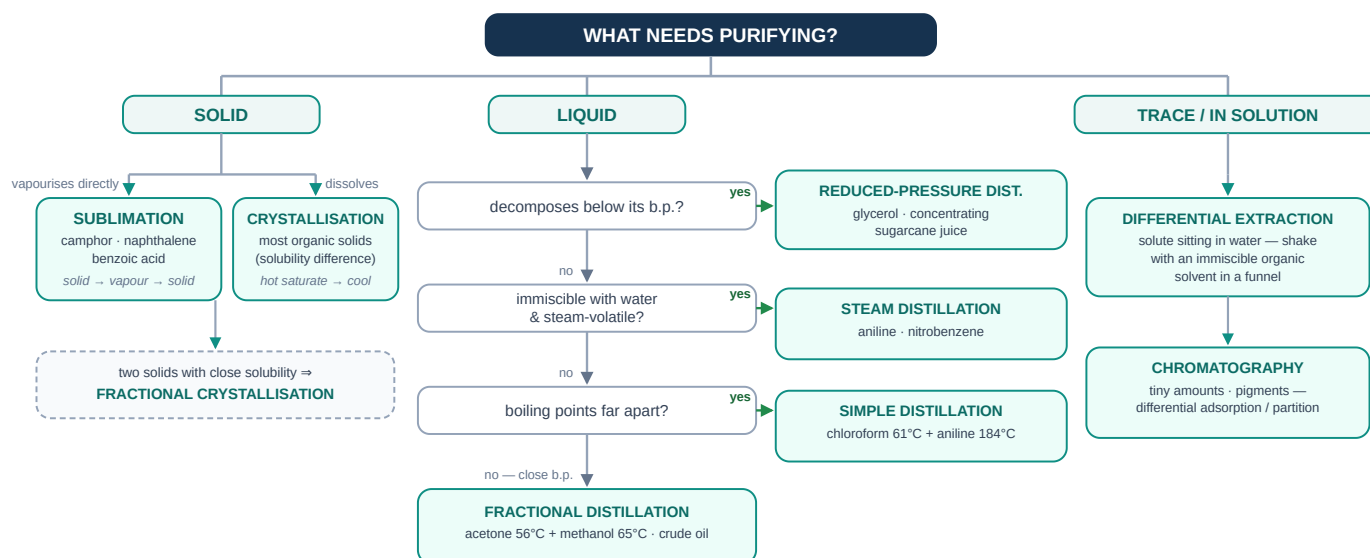
pure solid ⇒ sharp, constant m.p.

pure liquid ⇒ constant b.p.

impurity ⇒ m.p. falls & broadens — the mixed-m.p. identity check

2 Which Purification Method? — Ask in Order

PURIFICATION

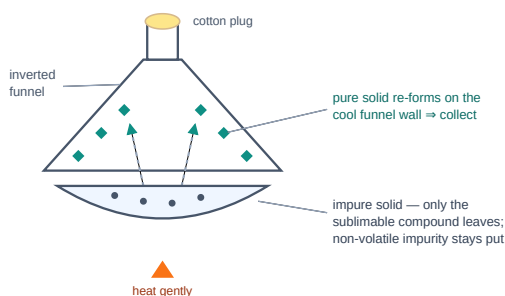


Left to right this is NCERT's own sequence — sublimation & crystallisation, the distillations, then extraction and chromatography. The examples are the ones examiners quote.

3 Sublimation & Crystallisation — NCERT's First Two Methods

PURIFICATION

SUBLIMATION — SOLID → VAPOUR → SOLID



camphor · naphthalene · benzoic acid · NH_4Cl

skips the liquid state entirely

CRYSTALLISATION — THE FOUR-STEP ROUTINE

- Dissolve** the impure solid in the **minimum volume of hot solvent** — a nearly saturated solution.
- Filter hot** — insoluble impurities stay on the paper.
- Cool slowly, undisturbed** — pure crystals grow; soluble impurities stay behind in the mother liquor.
- Filter, wash, dry** the crystals — then confirm purity by a **sharp melting point** (Section 1).



Solvent rule: the compound must be **sparingly soluble cold** but **freely soluble hot**; impurities must stay either wholly soluble or wholly insoluble.

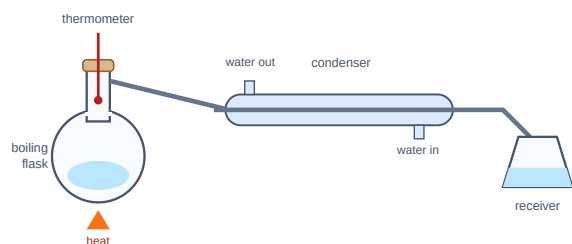
coloured impurity ⇒ boil with animal charcoal, filter hot

two solids, close solubility ⇒ repeat = **FRACTIONAL** crystallisation

4 Distillation — Four Set-ups, Four Situations

PURIFICATION

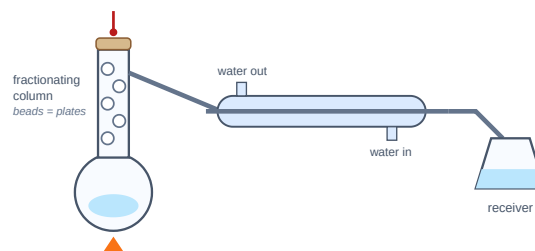
SIMPLE DISTILLATION



$\Delta b.p.$ large · neither decomposes

chloroform 61 °C + aniline 184 °C

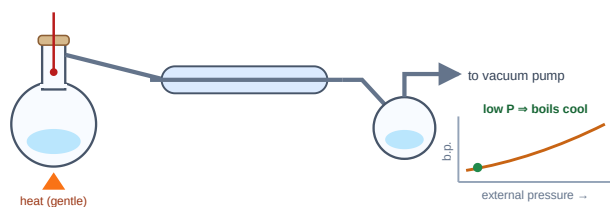
FRACTIONAL DISTILLATION



$b.p.$ close together

acetone 56 °C + methanol 65 °C · crude oil

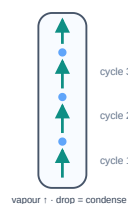
DISTILLATION UNDER REDUCED PRESSURE



decomposes below its $b.p.$

glycerol · concentrating sugarcane juice

WHY A COLUMN WORKS — THEORETICAL PLATES



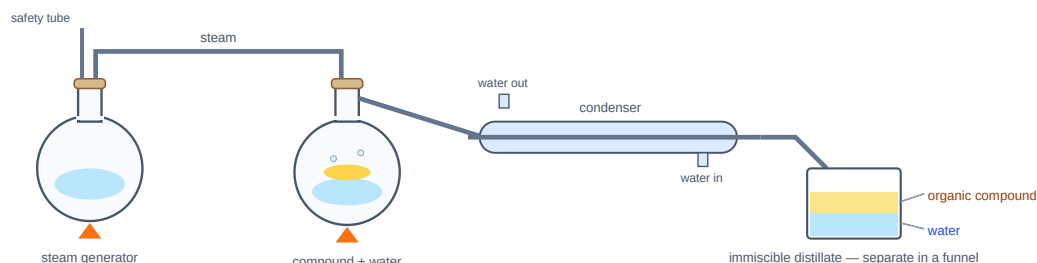
1 vapourise–condense cycle = 1 theoretical plate.

Every cycle leaves the vapour richer in the **more volatile** liquid, so a tall column separates even close-boiling mixtures.

Azeotrope = constant-boiling mixture — it distils with **unchanged composition**. Ethanol–water at **95.6 % ethanol**: **no column, however long**, can separate it.

5 Steam Distillation — the Compound Boils Out Below 100 °C

PURIFICATION



Each immiscible liquid exerts its **full** vapour pressure — the sum reaches 1 atm **below 100 °C**, so the compound distils without decomposing.

CONDITIONS

- ✓ steam-volatile
- ✓ immiscible with water
- ✓ does not react with water
- ✓ fair vapour pressure near 100 °C

WHY IT BOILS EARLY

$$p_A + p_B = p_{\text{atm}} \Rightarrow \text{boils below } 100\text{ }^{\circ}\text{C}$$

DISTILLATE RATIO

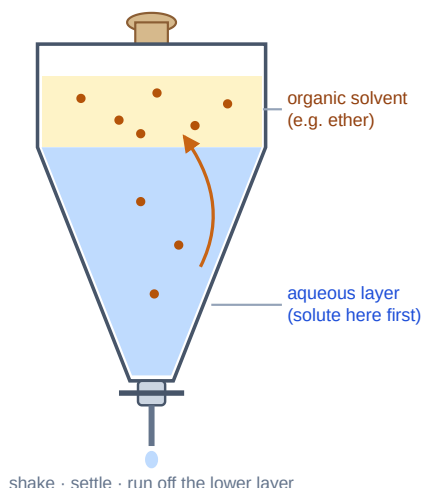
$$\frac{w_A}{w_B} = \frac{p_A M_A}{p_B M_B}$$

SYMBOLS

A = organic · B = water · p = vapour pressure
 M = molar mass · w = mass in distillate

6 Differential Extraction — Pulling a Compound Out of Water

PURIFICATION



THE THREE MOVES

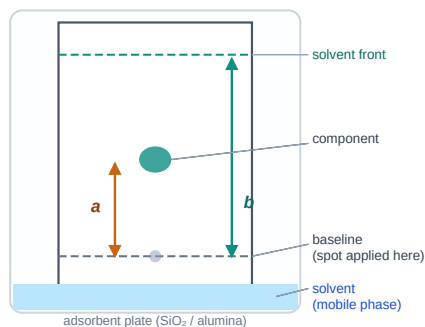
- 1 Add an **immiscible organic solvent** (e.g. ether) to the aqueous solution in a separating funnel — the compound must dissolve far better in it than in water.
- 2 **Shake gently, then let the layers settle** — the organic compound crosses into the solvent it prefers.
- 3 **Run off the lower layer**, keep the organic layer, and **evaporate the solvent** — the compound is left behind.

i Teacher's note: repeat with **fresh portions of solvent** to recover more of the compound; when it is only sparingly soluble in the solvent, NCERT prescribes **continuous extraction** instead.

7 Chromatography & the R_f Value

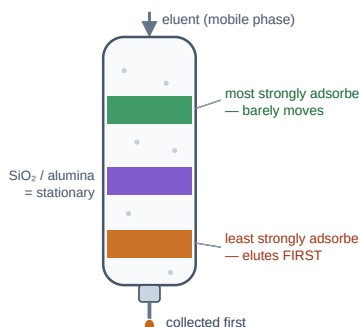
PURIFICATION

THIN-LAYER (TLC) — MEASURING R_f



$$R_f = \frac{\text{distance moved by component}}{\text{distance moved by solvent front}} = \frac{a}{b}$$

COLUMN — WHO LEAVES FIRST?



HOW TO READ IT

$0 \leq R_f \leq 1$ — dimensionless

both distances measured from the same baseline

more polar solute on a polar phase $\Rightarrow$ sticks harder $\Rightarrow$ lower R_f

weakest-held component leaves the column first

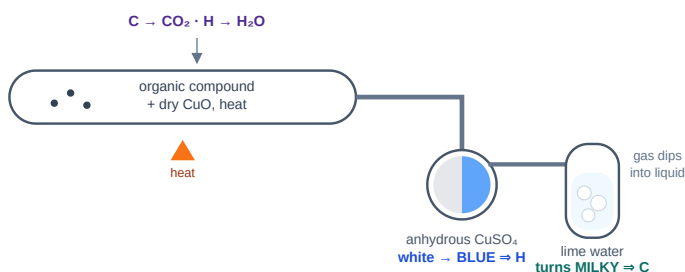
ADSORPTION
stationary = solid
 $\rightarrow$ column, TLC

PARTITION
stationary = liquid
 $\rightarrow$ paper chromatography

i Paper chromatography is PARTITION, not adsorption — the stationary phase is the **water held in the paper's fibres**, not the cellulose itself.

8 Detecting Carbon & Hydrogen — the CuO Combustion

DETECTION



CARBON



HYDROGEN

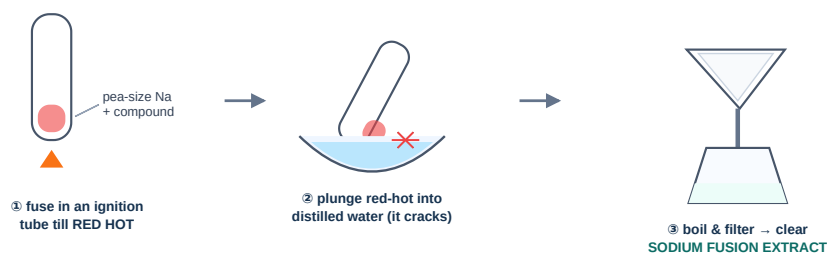


i Gases pass over **$CuSO_4$ first** — bubbled through lime water first, they would carry moisture on and turn $CuSO_4$ blue even with no H in the compound.

i **C & H never come from the Lassaigne extract** — this combustion is their own test.

9 Lassaing's Sodium Fusion — Making N, S, X Ionic

DETECTION



WHAT THE FUSION MAKES (ALL IONIC)

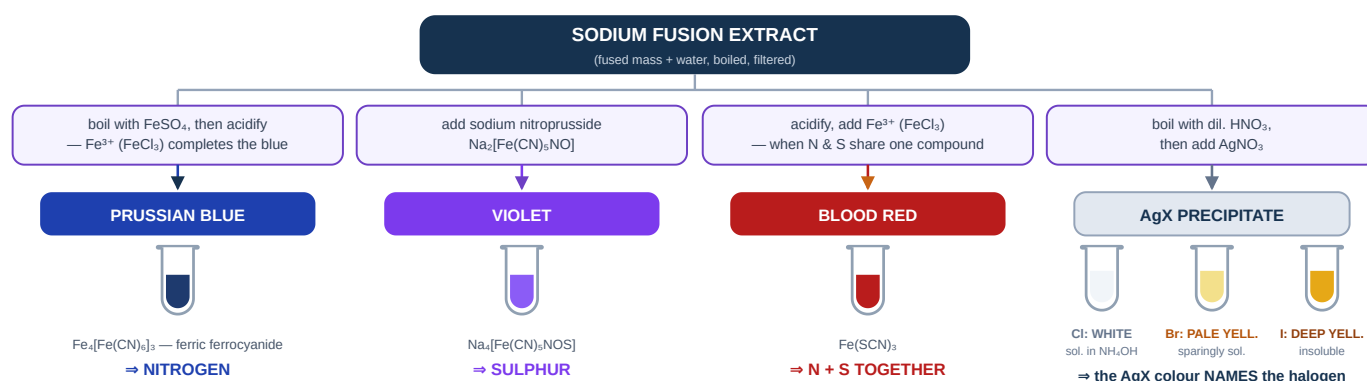


NaSCN forms when N & S sit in one compound · X = Cl, Br, I

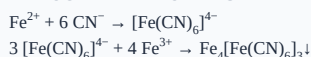
Why sodium? Covalently bound N, S and X give **no ions** in water, so no ionic test could see them — fusion converts them into **ionic sodium salts**, and every test on the next page works on that extract.

10 One Extract, Four Tests — the Complete Decision Map

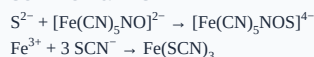
DETECTION



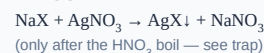
NITROGEN — THE CHEMISTRY



SULPHUR & N+S



HALOGEN



1 Boil with dil. HNO_3 BEFORE AgNO_3 — it destroys NaCN & Na_2S , which would otherwise give white AgCN / black Ag_2S = a **false halogen test**.

2 With excess Na, $\text{NaSCN} \rightarrow \text{NaCN} + \text{Na}_2\text{S}$ — N & S then answer **separately** (Prussian blue + violet), and the blood-red test **fails**.

PHOSPHORUS

oxidise (Na_2O_2) → phosphate; boil with conc. HNO_3 , add **ammonium molybdate** → **canary-yellow ppt** (ammonium phosphomolybdate)

BEILSTEIN TEST

Cu wire + compound in flame → **green-blue flame** = halogen, but **preliminary only** — urea & thiourea give a false positive

Prussian blue
⇒ N

violet (nitroprusside)
⇒ S

blood red $\text{Fe}(\text{SCN})_3$
⇒ N + S

black PbS (lead acetate)
⇒ S

white AgCl , sol. NH_4OH
⇒ Cl

pale yellow AgBr
⇒ Br

deep yellow AgI
⇒ I

green flame (Beilstein)
⇒ halogen, preliminary

The colour is the answer — learn the swatch, not the sentence.

11 One Formula Runs Every Estimation

ESTIMATION

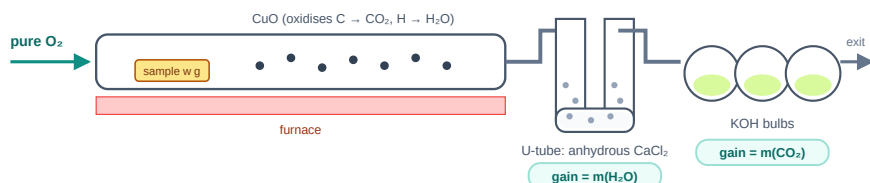
$$\% \text{ element} = \frac{\text{mass of the element in 1 mol of product}}{\text{molar mass of that product}} \times \frac{\text{mass of product collected}}{\text{mass of compound taken, } w} \times 100$$

GRAVIMETRIC FACTOR — fixed, pure chemistry FROM THE EXPERIMENT — your two weighings

Learn this once — every estimation that follows is this very line with a different collected product.

12 Carbon & Hydrogen — Liebig's Combustion

ESTIMATION



CARBON

$$\%C = \frac{12}{44} \times \frac{m_{\text{CO}_2}}{w} \times 100$$

HYDROGEN

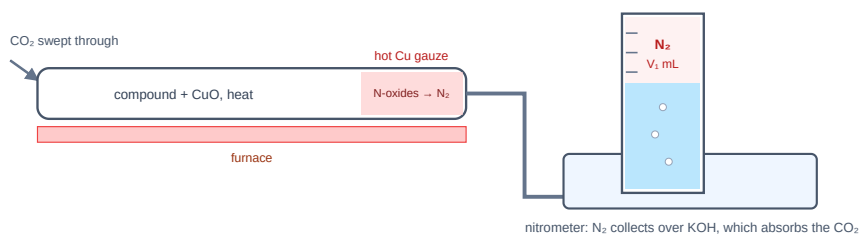
$$\%H = \frac{2}{18} \times \frac{m_{\text{H}_2\text{O}}}{w} \times 100$$

2/18 = 1/9 · Mg(ClO₄)₂ also traps water

Order is critical: the CaCl₂ tube goes **BEFORE** the KOH bulbs — KOH absorbs water as well as CO₂, so swapping them ruins **both** readings.

13 Nitrogen — Dumas' Method

ESTIMATION



Heated with CuO in a CO₂ atmosphere — hot copper gauze turns every oxide of nitrogen back into N₂; KOH absorbs the CO₂ carrier, so **only** N₂ is collected.

NITROGEN (DUMAS)

$$\%N = \frac{28}{22400} \times \frac{V_{\text{STP}} \text{ (mL)}}{w} \times 100$$

REDUCE THE VOLUME TO STP

$$\frac{P_1 V_1}{T_1} = \frac{760 \times V_{\text{STP}}}{273}$$

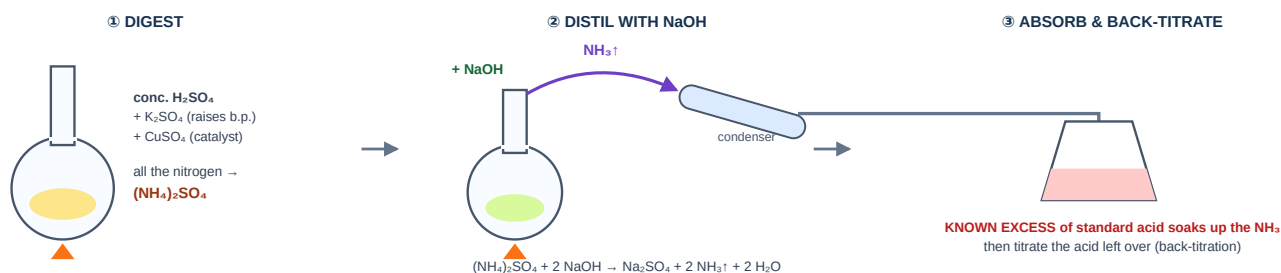
$P_1 = P_{\text{atm}}$ — aqueous tension (gas is moist)

28 g N₂ ≡ 22400 mL at STP

works for every N-compound — ring-N, nitro, azo included

14 Nitrogen — Kjeldahl's Method (Three Steps)

ESTIMATION



DIRECT FORM

$$\%N = \frac{1.4 \times N_{\text{acid}} \times V_{\text{acid}} \text{ (mL)}}{w}$$

= meq of acid actually used by NH₃

BACK-TITRATION FORM

$$\%N = \frac{1.4 (N_1 V_1 - N_2 V_2)}{w}$$

$N_1 V_1$ = acid taken · $N_2 V_2$ = alkali for the excess

SHORTCUTS

$$1.4 = \frac{14 \times 100}{1000} \text{ — nothing to memorise}$$

HCl: N = M H₂SO₄ (dibasic): N = 2M

Kjeldahl fails for nitrogen in a **ring** (pyridine, quinoline) and in **nitro / azo** groups — that N never becomes (NH₄)₂SO₄. **Dumas works for all of them.**

15 Halogens, Sulphur & Phosphorus — Carius Method

ESTIMATION

FURNACE

compound (w g) + fuming HNO_3 + AgNO_3
SEALED hard-glass tube, heated strongly

→

AgX ↓ precipitates

filter · wash · dry · WEIGH

HALOGEN

$\text{X} \rightarrow \text{AgX}$ directly — weigh it

SULPHUR

$\text{S} \rightarrow \text{H}_2\text{SO}_4 \rightarrow \text{BaCl}_2 \rightarrow \text{BaSO}_4 \downarrow$ (weighed)

PHOSPHORUS

$\text{P} \rightarrow \text{H}_3\text{PO}_4 \rightarrow \text{MgNH}_4\text{PO}_4 \rightarrow \text{ignite} \rightarrow \text{Mg}_2\text{P}_2\text{O}_7$ (weighed)

C weighed as CO_2 $\frac{12}{44}$	H weighed as H_2O $\frac{2}{18}$	N as N_2 at STP (mL) $\frac{28}{22400}$
Cl weighed as AgCl $\frac{35.5}{143.5}$	Br weighed as AgBr $\frac{80}{188}$	I weighed as AgI $\frac{127}{235}$
S weighed as BaSO_4 $\frac{32}{233}$	P weighed as $\text{Mg}_2\text{P}_2\text{O}_7$ $\frac{62}{222}$	O as CO_2 (direct method) $\frac{32}{88}$

Every estimation on one row of cards: $\%E = \text{factor} \times (m_{\text{product}} / w) \times 100$ — the factor is just (mass of E in 1 mol of the weighed product) ÷ (molar mass of that product).

16 Oxygen — by Difference, or Directly

ESTIMATION

BY DIFFERENCE — THE USUAL ROUTE

$\%O = 100 - (\%C + \%H + \%N + \%S + \%X + \dots)$

DIRECT METHOD — EVERY O ENDS UP IN A CO_2

compound heated in a stream of N_2 → over red-hot coke (1373 K): all its O → CO → I_2O_5 oxidises it: $5\text{CO} + \text{I}_2\text{O}_5 \rightarrow 5\text{CO}_2 + \text{I}_2$ → weigh the CO_2 or titrate the I_2

VIA THE CO_2 COLLECTED

$$\%O = \frac{32}{88} \times \frac{m_{\text{CO}_2}}{w} \times 100$$

VIA THE I_2 LIBERATED

$$\%O = \frac{80}{254} \times \frac{m_{\text{I}_2}}{w} \times 100$$

1 $\text{I}_2 \equiv 5\text{CO} \equiv 5\text{O}$ from the compound (80 g)

! Why 32/88 and not 32/44? Each CO_2 finally collected carries only **one** oxygen that came from the compound — the second O comes from I_2O_5 . So the factor is 32/88 = 16/44. **The single most-missed point in this chapter.**

17 Molecular Mass — Three Classical Routes

FORMULA

VICTOR MEYER — volatile compounds

$$M = 2 \times V.D. = \frac{w \times 22400}{V_{\text{STP}} (\text{mL})}$$

vapourise w g, measure the air it displaces

SILVER SALT — for acids

$-\text{COOH} \rightarrow -\text{COOAg}$: swap H (1) for Ag (108); ignite → pure Ag

$$E = \frac{108 \times w_{\text{salt}}}{w_{\text{Ag}}} - 107, \quad M = E \times \text{basicity}$$

CHLOROPLATINATE — for bases

base B → $\text{B}_2 \cdot \text{H}_2\text{PtCl}_6$ ($M = 2E + 410$); ignite → Pt (195)

$$E = \frac{195 \times w_{\text{salt}}}{2 \times w_{\text{Pt}}} - 205, \quad M = E \times \text{acidity}$$

18 Empirical → Molecular Formula & Degree of Unsaturation

FORMULA

FROM % TO FORMULA — FOUR MECHANICAL STEPS

% of each element → ÷ atomic mass = relative moles → ÷ the smallest → × small integer → whole numbers → **EMPIRICAL FORMULA**

molecular formula = $n \times$ (empirical formula), $n = \frac{\text{molecular mass}}{\text{empirical formula mass}}$

DEGREE OF UNSATURATION (DoU)

$$\text{DoU} = \frac{2C + 2 + N - H - X}{2}$$

ring or $\text{C}=\text{C} = 1$ $\text{C}\equiv\text{C} = 2$

O appears nowhere in it

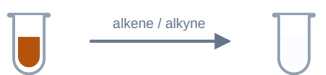
benzene ring = 4 (3 π + 1 ring) — fastest structural clue

Work the chain in this order: **% composition** → **molecular mass** → **empirical formula** → **molecular formula** → **DoU** — and let the DoU tell you which functional-group tests to run next.

19 Structural ID — Finding the Unsaturation

STRUCTURAL ID

BROMINE WATER



orange-brown

colourless

Br_2 adds across the multiple bond, so the colour is discharged $\Rightarrow$ unsaturation present.

BAEYER'S REAGENT

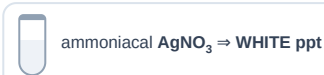


pink KMnO_4

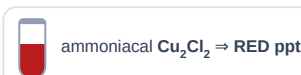
colourless + brown MnO_2

Cold, dilute, alkaline KMnO_4 — the alkene becomes a vicinal diol.

TERMINAL ALKYNE ($-\text{C}\equiv\text{C}-\text{H}$)



ammoniacal $\text{AgNO}_3 \Rightarrow$ WHITE ppt



ammoniacal $\text{Cu}_2\text{Cl}_2 \Rightarrow$ RED ppt

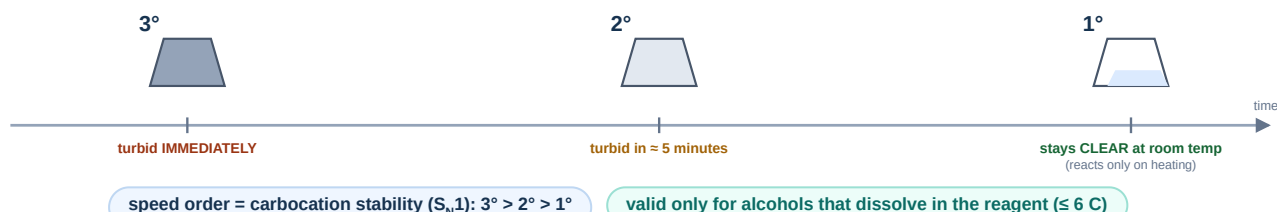
An internal alkyne gives **no precipitate** — this is how the triple bond is **located**.

! **Baeyer's is really an "easily oxidised?" test** — aldehydes and other readily oxidised groups also discharge the pink. Never read it as $\text{C}=\text{C}$ -only evidence.

20 Alcohols — Telling 1°, 2°, 3° Apart

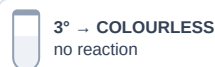
STRUCTURAL ID

LUCAS TEST — conc. HCl + anhydrous ZnCl_2



VICTOR MEYER COLOUR TEST

$\text{R}-\text{OH} \xrightarrow{\text{HI}} \text{R}-\text{I} \xrightarrow{\text{AgNO}_2} \text{R}-\text{NO}_2 \xrightarrow{\text{HNO}_2} \text{HNO}_2$, then $\text{NaOH} \rightarrow$ colour

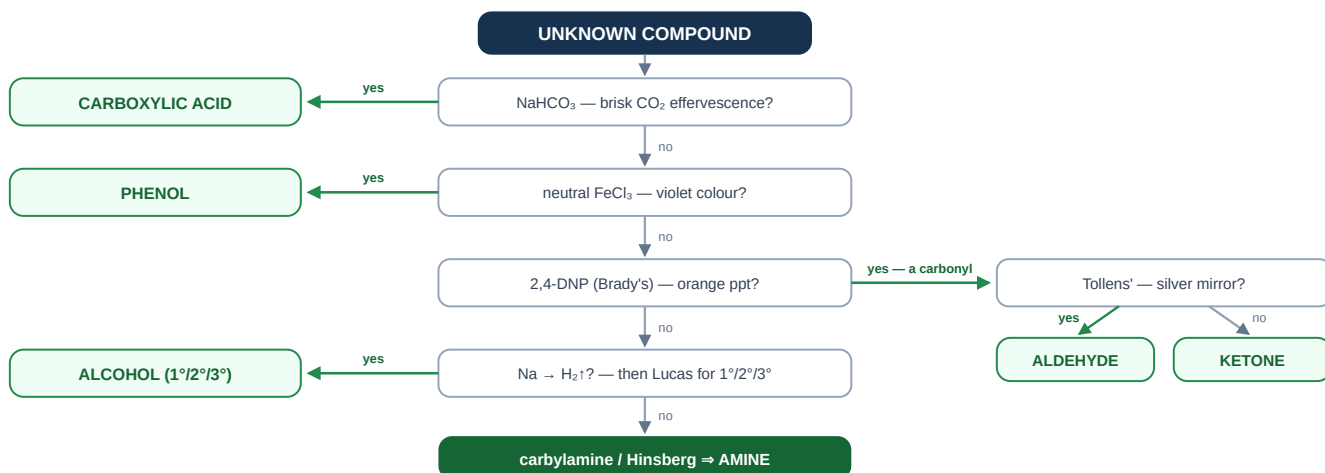


SODIUM METAL — SUPPORTING ONLY

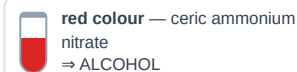
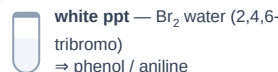
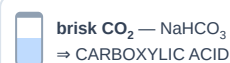
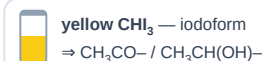
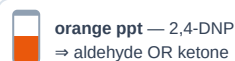
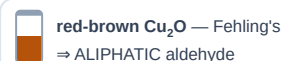
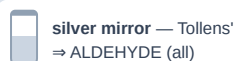
Brisk H_2 effervescence shows an $-\text{OH}...$ but **carboxylic acids and phenols bubble too** — treat it as supporting evidence, never as proof of an alcohol.

21 Carbonyl, Acid & Phenol — the Unknown-Compound Tree

STRUCTURAL ID



A working order: acid $\rightarrow$ phenol $\rightarrow$ carbonyl $\rightarrow$ alcohol $\rightarrow$ amine — one cheap test per rung.



iodoform — WHO SAYS YES? ($I_2 + NaOH \rightarrow$ yellow $CHI_3 \downarrow$)

POSITIVE ✓

ethanal · acetone · **every methyl ketone** · ethanol
· propan-2-ol

NEGATIVE ✗

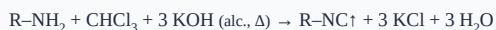
methanol · propanal · benzaldehyde · pentan-3-one

! **Classic JEE trap:** benzaldehyde gives the Tollens' mirror but does **NOT** respond to Fehling's — Fehling's is for **aliphatic** aldehydes only.

22 Amines & Nitro Compounds

STRUCTURAL ID

CARBYLAMINE (ISOCYANIDE) TEST — 1° ONLY



foul-smelling isocyanide

given by 1° amines only — aliphatic and aromatic

2° & 3°: silent

HINSBERG — $C_6H_5SO_2Cl$, THEN aq. KOH

1° sulphonamide with an acidic N-H

dissolves in KOH — clear

2° sulphonamide with no N-H

insoluble in KOH

3° no reaction

amine unchanged

NITROUS ACID — $NaNO_2 +$ dil. HCl AT 0–5 °C



1° aliphatic
brisk $N_2 \uparrow$ + alcohol



1° aromatic → stable diazonium salt — β-naphthol → **orange-red** azo dye



2° (both)
yellow oily N-nitrosoamine



3° aliphatic
just dissolves as a salt

why 0–5 °C? — any warmer and the diazonium salt decomposes to phenol + N_2

N Nitro group (Mulliken–Barker): reduce with Zn dust + $NH_4Cl \rightarrow$ a hydroxylamine, which then gives a **silver mirror** with Tollens' reagent.

23 Distinguishing Common Pairs — One Decisive Test Each

STRUCTURAL ID

PAIR	ONE DECISIVE TEST
carboxylic acid / phenol	$NaHCO_3$ — only the acid gives brisk CO_2
phenol / alcohol	neutral $FeCl_3$ — only phenol turns violet
aldehyde / ketone	Tollens' — only the aldehyde silvers the tube
acetaldehyde / benzaldehyde	Fehling's — only the aliphatic one gives red Cu_2O
acetone / propanal	iodoform — only acetone (CH_3CO-) gives CHI_3
ethanol / methanol	iodoform — only ethanol responds
1° / 2° / 3° alcohol	Lucas (now / ~5 min / clear), or Victor Meyer (red / blue / colourless)
1° / 2° / 3° amine	Hinsberg's, or carbylamine (answers for 1° only)
aniline / phenol	carbylamine → aniline ; $FeCl_3$ violet → phenol
terminal / internal alkyne	ammoniacal $AgNO_3$ — only the terminal one precipitates
alkene / alkane	bromine water — only the alkene decolourises it

24 All Formulae at a Glance

REVISION

QUANTITY	FORMULA	QUANTITY	FORMULA
R_f	$\frac{\text{distance by component}}{\text{distance by solvent front}}$	%S	$\frac{32}{233} \times \frac{m_{\text{BaSO}_4}}{w} \times 100$
steam distillation	$\frac{w_A}{w_B} = \frac{p_A M_A}{p_B M_B}$	%P	$\frac{62}{222} \times \frac{m_{\text{Mg}_2\text{P}_2\text{O}_7}}{w} \times 100$
%C	$\frac{12}{44} \times \frac{m_{\text{CO}_2}}{w} \times 100$	%O	$100 - \Sigma(\text{others}); \text{ direct: } \frac{32}{88} \times \frac{m_{\text{CO}_2}}{w} \times 100$
%H	$\frac{2}{18} \times \frac{m_{\text{H}_2\text{O}}}{w} \times 100$	molecular formula	$n = \frac{\text{molecular mass}}{\text{empirical formula mass}}$
%N (Dumas)	$\frac{28}{22400} \times \frac{V_{\text{STP}}}{w} \times 100$	DoU	$\frac{2C + 2 + N - H - X}{2}$
%N (Kjeldahl)	$\frac{1.4 N V}{w}$; back-titration $\frac{1.4(N_1 V_1 - N_2 V_2)}{w}$	molecular mass	$M = 2 \times \text{VD} = \frac{w \times 22400}{V_{\text{STP}}}$
%Cl, %Br, %I	$\frac{35.5}{143.5}, \frac{80}{188}, \frac{127}{235} \times \frac{m_{\text{AgX}}}{w} \times 100$	silver salt (acid)	$E = \frac{108 w_{\text{salt}}}{w_{\text{Ag}}} - 107; M = E \times \text{basicity}$
		platinichloride (base)	$E = \frac{195 w_{\text{salt}}}{2 w_{\text{Pt}}} - 205; M = E \times \text{acidity}$

25 Ten Traps That Decide the Question

REVISION

- In Liebig's method the **CaCl₂ tube goes BEFORE the KOH bulbs** — never the other way round.
- Boil the extract with dil. HNO₃ before AgNO₃** — else NaCN and Na₂S give a false halogen precipitate.
- With **excess Na**, NaSCN → NaCN + Na₂S ⇒ the N and S tests appear **separately**, not as blood-red.
- Kjeldahl fails** for ring-N (pyridine), nitro and azo compounds; **Dumas works for all**.
- Direct oxygen factor is **32/88, not 32/44** — only one O in each collected CO₂ came from the compound.
- Benzaldehyde**: Tollens' **yes**, Fehling's **no**.
- Iodoform needs **CH₃CO-** or **CH₃CH(OH)-**: ethanol ✓ methanol ✗ · acetone ✓ propanal ✗.
- The **Beilstein** green flame is **preliminary only** — urea and thiourea give a false positive.
- An **azeotrope** (ethanol–water, 95.6 %) cannot be separated by fractional distillation, however long the column.
- Dumas collects moist N₂** (over liquid) — take $P_1 = P_{\text{atm}} - \text{aqueous tension}$ **before** reducing the volume to STP.